

ARTIFICIAL INTELLIGENCE

BS Computer Science — 6th Semester

Capital University of Science & Technology, Islamabad

Week 1 Report

AI-Based Long-Term Investment Advisory & Portfolio Optimization System

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1. Problem Definition

This project aims to design an intelligent investment advisory system that helps long-term investors make portfolio allocation decisions. The system uses heuristic evaluation, local search algorithms, and explainable recommendations all techniques covered in the AI course syllabus.

1.1 Problem Statement

Individual investors lack access to intelligent, personalized advisory tools that can analyze multiple stocks against their risk profile, investment duration, and return expectations. This system addresses that gap by providing explainable, optimized portfolio recommendations using classical AI techniques.

1.2 System Objectives

- Model investment decisions as an AI optimization problem
- Define and implement a heuristic scoring system for stock evaluation
- Implement portfolio optimization using Hill Climbing and Simulated Annealing
- Generate explainable, plain-English investment recommendations
- Visualize portfolio allocations, risk-return tradeoffs, and projected growth

1.4 System Constraints

The system is not expected to perform real-time trading, connect to brokerage APIs, use deep learning, or predict exact future prices. The focus is on intelligent recommendation and portfolio optimization using classical AI techniques covered in the course.

2. Dataset Collection

Data was collected from Yahoo Finance using the yfinance Python library (v1.4.1), accessing PSX-listed securities with the .KA ticker suffix (e.g. MCB.KA, FFC.KA).

2.1 Collection Method

A custom script (collect_data.py) fetches 5 years of daily OHLCV data for each stock using yf.download(). Dividend yield is fetched separately via yf.Ticker().info. Where Yahoo Finance returns zero for dividend yield — which is common for PSX tickers — values were substituted from PSX annual report data.

2.2 Target Stocks

15 PSX-listed stocks were selected across multiple sectors: MCB, ENGRO, HBL, OGDC, PPL, LUCK, PSO, UBL, NESTLE, SEARL, HUBC, KAPCO, FFC, EFERT, TRG.

Time range: June 2020 – June 2025 (5-year window). After aggregation, the final stocks.csv contains 15 records with 11 features each.

3. Stock Features Identified

3.1 Feature Definitions

The following features were identified as inputs to the heuristic scoring system:

Feature	Description	Calculation
Avg Annual Growth (%)	Long-term return trend over 5 years	CAGR from historical prices
Volatility	Risk level based on price fluctuation	Std dev of daily returns × $\sqrt{252} \times 100$
Dividend Yield (%)	Passive income potential	Annual dividend / stock price
Sector	Industry classification	PSX sector listing
Historical Stability	Long-term consistency score (0–100)	Composite stability metric
Risk Score (0–100)	Overall risk level	Derived from volatility and beta

3.2 Sample Data (5 Stocks)

The table below shows collected data for 5 of the 15 target stocks. This represents the format of the full stocks.csv dataset.

Ticker	Company	Sector	Price (PKR)	Avg Growth (%)	Volatility (%)	Vol. Label	Div. Yield (%)	Avg Daily Vol.
MCB	MCB Bank Limited	Banking	412.62	38.18	25.38	Medium	8.5	588,046
ENGRO	Engro Corporation	Fertilizer	485.38	31.71	24.85	Medium	5.2	723,040
HBL	Habib Bank Limited	Banking	286.06	26.78	32.64	Medium	6.0	2,242,058
OGDC	Oil & Gas Dev. Co.	Energy	324.56	37.18	35.05	High	7.8	5,944,377
PPL	Pakistan Petroleum	Energy	232.78	24.72	38.52	High	6.5	8,281,448

4. Heuristic Design Draft

The heuristic scoring system evaluates each stock against a weighted set of criteria derived from the investor's profile. Each stock receives a score between 0 and 100, which determines its suitability for a portfolio.

4.1 Scoring Weights

Factor	Weight
Long-Term Growth	+30
Sector Diversification	+20
Low Volatility	+20
Dividend Stability	+15
Momentum Trend	+15

4.2 Scoring Formula

Score = $\sum (\text{factor_value} \times \text{weight})$, adjusted by investor risk profile.

For conservative profiles, the Low Volatility weight increases to +30 while Long-Term Growth decreases to +20. For aggressive profiles, the inverse adjustment is applied.

4.3 Explainability

For each recommended stock, the system generates a plain-English explanation. For example:

"MCB was selected because of: strong historical stability, low volatility matching your medium risk profile, and consistent 8.5% dividend yield."

5. Initial UI Mockup

The application has been built as a Streamlit-based web application with four pages, each corresponding to a weekly deliverable. All four pages are complete with placeholder data for the Week 1 submission.

5.1 Technology Stack

Component	Choice
Framework	Streamlit (Python)
Charts	Plotly
Theme	Dark mode (#0f1117 background)
Fonts	IBM Plex Sans (headings), Inter (body)

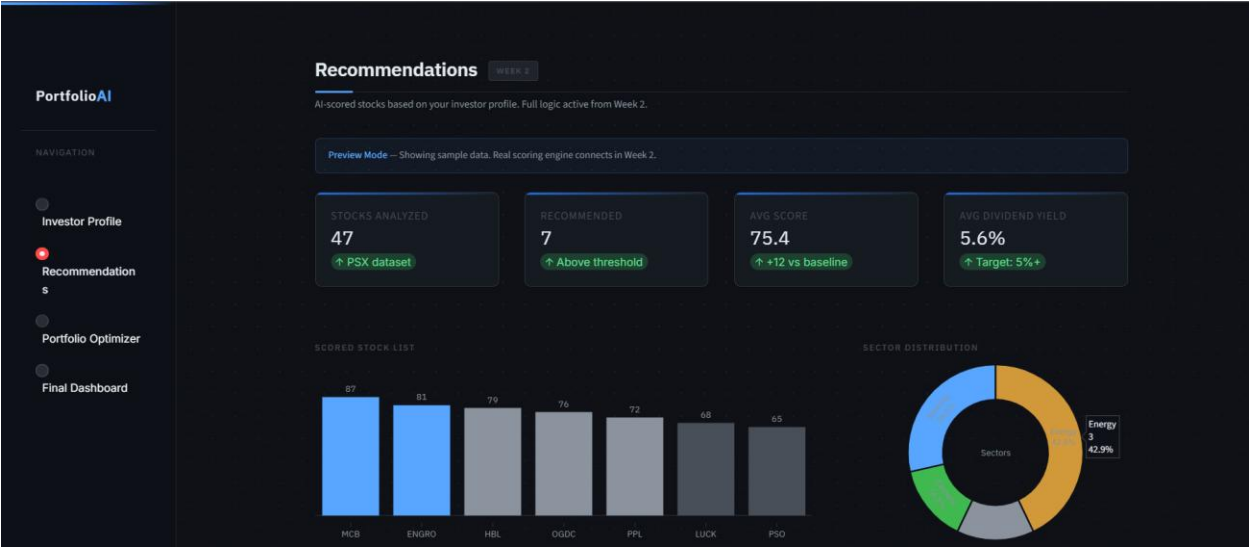
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5.2 Application Pages

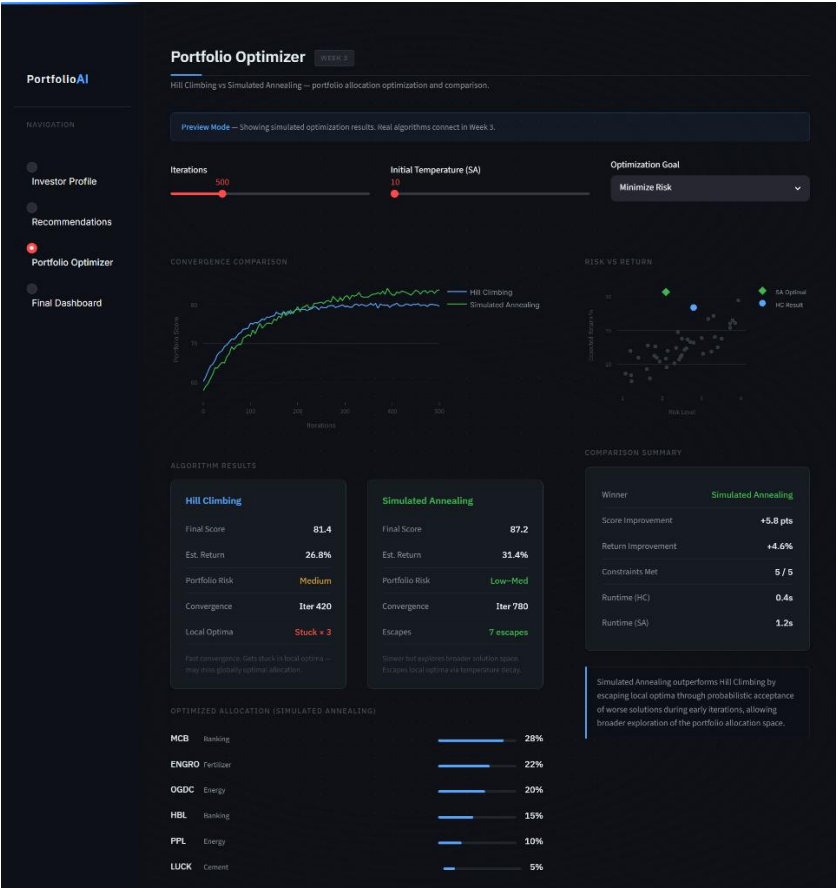
- Page 1 — Investor Profile: Input form collecting investment amount, duration, risk appetite, target return, preferred sectors, and excluded sectors. A live profile summary card updates as the user fills the form.
- Page 2 — Recommendations (Week 2): AI-scored stock list with bar chart, expandable stock cards, explainability blocks, and sector distribution pie chart.
- Page 3 — Portfolio Optimizer (Week 3): Side-by-side comparison of Hill Climbing vs Simulated Annealing with convergence curves, allocation breakdown, and risk vs return scatter plot.
- Page 4 — Final Dashboard (Week 4): Complete portfolio overview with projected growth chart, allocation pie chart, full explainability report, and export functionality.

Screenshots of all four pages of the running application are provided bellow.

Investor Profile Page



Recommendations Page



Portfolio Optimizer Page

